

Module 53 - Fetch API

1. What is Fetch API?

- Fetch API is a built-in JavaScript API used to send HTTP requests and receive data from a server.
- It returns a Promise that resolves to the response of the request.
- By default, fetch() uses the GET HTTP method.

Example Code:

I. Fetch API using .then() and .catch()

```
// Sends a GET request to the API.
fetch("https://fakestoreapi.com/users/abcd")
.then((response) => {
    // Checks whether the HTTP response was successful.
    if(!response.ok){
        throw new Error("Data Not Found");
    }
    else{
        // Converts the response body from JSON into JavaScript data.
        return response.json();
    }
})
.then((data) => {
    // Receives and displays the converted data.
    console.log(data);
})
.catch((error) => {
    // Handles any error from the request or Promise chain.
    console.log(error.message);
})
```

II. Fetch API using async & await

```
async function fetchData(){
  try{
    // Sends a GET request and waits for the response.
    let response = await fetch("https://fakestoreapi.com/users");
    // Checks whether the HTTP response was successful.
    if(!response.ok){
      throw new Error("Data Not Found");
    }
    else{
      // Converts the response body from JSON into JavaScript data.
      let data = await response.json();
      // Displays the first user from the returned array.
      console.log(data[0]);
    }
  }
  catch(error){
    // Handles errors from fetch or the thrown error.
    console.log(error.message);
  }
}

// Calls the async function.
fetchData();
```

2. HTTP Methods in REST API

- **GET** → Used to retrieve data from a server.
- **POST** → Used to create/send new data on a server.
- **PUT** → Used to replace or update existing data on a server
- **DELETE** → Used to delete data from a server.
- **HEAD** → Used to get response headers without the response body.
- **OPTIONS** → Used to check which HTTP methods or options are supported by a server.
- **PATCH** → Used to partially update existing data on a server.
- **TRACE** → Used to diagnose the path of a request between the client and server.
- **CONNECT** → Used to establish a network connection to a server, commonly for a proxy tunnel.